23 It would take 120 minutes to fill a swimming pool using water from 5 taps.

(a) How many minutes will it take to fill the pool if only 3 of the taps are used?

Taps × minutes stays constant: 5 × 120 = 600.

1 mark
Use inverse proportion: 120 × 5 ÷ 3

For 3 taps: 600 ÷ 3 = 200 minutes.

200 ..... minutes

1 mark
Correct time

(2)

(b) State one assumption you made in working out your answer to part (a).

All taps supply water at the same constant rate.

1 mark
State the rate assumption

(1)

(Total for Question 23 is 3 marks)

24 A plane travels at a speed of 213 miles per hour.

(a) Work out an estimate for the number of seconds the plane takes to travel 1 mile.

..... seconds

(3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

.....

.....

(1)

(Total for Question 24 is 4 marks)

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